Contraction is height-preserving admissible in G3c

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Editorial note. The proof cites three pieces of machinery by label: the inversion lemma (published as corpus card X2-proof-theory-sequent-calculus-g3c-invert), the quantifier inversion lemma, and the substitution regularity lemma; all three are proved in the source unit and are declared as prerequisites and as navigation references.

Editorial scope. One original proposition of the unit's structural admissibility section, counted once. The counted proposition and its author proof are retained in full.

Allowed standard prerequisites (editorial).

- The calculus G3c and the notion of height-preserving admissibility.
- Height-preserving invertibility of the G3c rules (the inversion lemma), used when the contracted formula is principal in the last inference.
- Height-preserving invertibility of the quantifier rules in the fresh-constant sense, and the regularity and substitution machinery for replacing a fresh constant by an arbitrary term without changing proof height.

Source reference. Original lemma G3c-invert: sources/olp-proof-theory-sequent-calculus/pinned-original-source.tex, source lines 963–963. The principal case applies the inversion lemma to the immediate subproofs. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. Published as corpus card X2-proof-theory-sequent-calculus-g3c-invert.

Source reference. Original lemma invert-quant: sources/olp-proof-theory-sequent-calculus/pinned-original-source.tex, source lines 1081–1081. The universal-right principal case applies part (1) of the quantifier inversion lemma to obtain the inverted premise at no greater height. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full.

Source reference. Original lemma subst-regular: sources/olp-proof-theory-sequent-calculus/pinned-original-source.tex, source lines 1700–1700. The universal-right principal case regularizes the inverted proof and applies the substitution lemma to replace the fresh constant while preserving regularity and height. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full.

Proposition 0.1. *The contraction rules and of G1c are height-preserving admissible in G3c.*

Proof. We give the argument for and at the same time. Suppose π is proof of $\Gamma \Rightarrow \Delta, A, A$ or of $A, A, \Gamma \Rightarrow \Delta$ in G3c. We have to show that there is also a G3c-proof π' of $\Gamma \Rightarrow \Delta, A$ or $A, \Gamma \Rightarrow \Delta$, respectively, with $\mathrm{ht}\,\pi' \leq \mathrm{ht}\,\pi$. We do this by induction on $n = \mathrm{ht}\,\pi$.

If $n = 0$, π is just an axiom. But if $\Gamma \Rightarrow \Delta, A, A$ is an axiom of G3c, so is $\Gamma \Rightarrow \Delta, A$: either some atomic C is element of both Γ and Δ, or A is atomic and element of Γ. The same reasoning applies if the end-sequent is $A, A, \Gamma \Rightarrow \Delta$.

If $n > 0$, π ends in some rule R. If A is not principal in this last inference, we can obtain proof π' as in the proof of G3c-invert by applying the inductive hypothesis to the immediate sub-proofs of π and applying the same rule R to the resulting proof(s). The inductive hypothesis is that if there is proof of height $k < n$ of either $\Pi \Rightarrow \Lambda, D, D$ or $D, D, \Pi \Rightarrow \Lambda$, then there is proof of height $\leq k$ of $\Pi \Rightarrow \Lambda, D$ or $D, \Pi \Rightarrow \Lambda$, respectively. (Note that the inductive hypothesis applies if the contexts or contracted formula are different from Γ, Δ, or A!)

For instance, suppose R is and π ends in

$$\dfrac{\begin{array}{c}\vdots\,\pi_1\\ \Gamma\Delta', B, A, A\end{array} \qquad \begin{array}{c}\vdots\,\pi_2\\ \Gamma\Delta', C, A, A\end{array}}{\Gamma\Delta', B \wedge C, A, A}$$

where $\Delta = \Delta', B \wedge C$. The inductive hypothesis yields proofs π_1' and π_2' of $\Gamma\Delta', B, A$ and $\Gamma\Delta', C, A$, respectively, with $\mathrm{ht}\,\pi_1' \leq \mathrm{ht}\,\pi_1$ and $\mathrm{ht}\,\pi_2' \leq \mathrm{ht}\,\pi_2$. The proof π' is:

$$\dfrac{\begin{array}{c}\vdots\,\pi_1'\\ \Gamma\Delta', B, A\end{array} \qquad \begin{array}{c}\vdots\,\pi_2'\\ \Gamma\Delta', C, A\end{array}}{\Gamma\Delta', B \wedge C, A}$$

If A is principal in the last inference, we apply the inversion lemma (G3c-invert) to the immediate sub-proofs of π, then the inductive hypothesis, and then rule R again. For instance, suppose π proves $\Gamma \Rightarrow \Delta, A, A$, $A \equiv (B \wedge C)$ and principal in the last inference. Then π looks as follows:

$$\dfrac{\begin{array}{c}\vdots\,\pi_1\\ \Gamma\Delta, B \wedge C, B\end{array} \qquad \begin{array}{c}\vdots\,\pi_2\\ \Gamma\Delta, B \wedge C, C\end{array}}{\Gamma\Delta, B \wedge C, B \wedge C}$$

Apply G3c-invert to π_1 to get proof π_1' of $\Gamma \Rightarrow \Delta, B, B$. (It also yields proof of $\Gamma \Rightarrow \Delta, C, B$, but we don't need that.) Similarly, by applying G3c-invert to π_2 we obtain proof π_2' of $\Gamma \Rightarrow \Delta, C, C$. As the inversion of is height-preserving, $\mathrm{ht}\,\pi_1' \leq \mathrm{ht}\,\pi_1 < \mathrm{ht}\,\pi$ and $\mathrm{ht}\,\pi_2' \leq \mathrm{ht}\,\pi_2 < \mathrm{ht}\,\pi$. Thus, the inductive hypothesis applies to π_1' and π_2': there are proofs π_1'' and π_2'' of $\Gamma \Rightarrow \Delta, B$ and $\Gamma \Rightarrow \Delta, C$, respectively, with $\mathrm{ht}\,\pi_1'' \leq \mathrm{ht}\,\pi_1' \leq \mathrm{ht}\,\pi_1$ and $\mathrm{ht}\,\pi_2'' \leq \mathrm{ht}\,\pi_2' \leq \mathrm{ht}\,\pi_2$. The proof π'

then is:

$$\dfrac{\begin{array}{c}\vdots\\ \pi_1''\\ \vdots\\ \Gamma\Delta, B\end{array} \qquad \begin{array}{c}\vdots\\ \pi_2''\\ \vdots\\ \Gamma\Delta, C\end{array}}{\Gamma\Delta, B \wedge C}$$

with $\operatorname{ht}\pi' \leq \operatorname{ht}\pi$.

We must take special care for the quantifier rules in the cases where the formula is principal.

Suppose $A \equiv \forall x\, B(x)$ is principal on the right. Then π ends in

$$\dfrac{\begin{array}{c}\vdots\\ \pi_1\\ \vdots\\ \Gamma\Delta, \forall x\, B(x), B(a)\end{array}}{\Gamma\Delta, \forall x\, B(x), \forall x\, B(x)}$$

By invert-quant , there is proof π_1' with height $\leq \operatorname{ht}\pi_1$ of $\Gamma\Delta, B(c), B(a)$ for any c not occuring in $\Gamma\Delta, \forall x\, B(x), B(a)$. We may assume that π_1' is regular; so by subst-regular the proof $\pi_1'[a/c]$ is proof of $\Gamma\Delta, B(a), B(a)$ also of height $\leq \operatorname{ht}\pi_1$. Now the inductive hypothesis applies and gives us proof π_1'' of $\Gamma\Delta, B(a)$. Apply the rule to get π':

$$\dfrac{\begin{array}{c}\vdots\\ \pi_1''\\ \vdots\\ \Gamma\Delta, B(a)\end{array}}{\Gamma\Delta, \forall x\, B(x)}$$

with $\operatorname{ht}\pi' \leq \operatorname{ht}\pi$.

Now suppose $A \equiv \exists x\, B(x)$ and principal on the right. Then π looks as follows:

$$\dfrac{\begin{array}{c}\vdots\\ \pi_1\\ \vdots\\ \Gamma\Delta, \exists x\, B(x), \exists x\, B(x), B(t)\end{array}}{\Gamma\Delta, \exists x\, B(x), \exists x\, B(x)}$$

The inductive hypothesis applies to π_1 and there is proofs π_1' of $\Gamma \Rightarrow \Delta, \exists x\, B(x), B$. (Note that no inversion lemma is needed here!) The proof π' then is:

$$\dfrac{\begin{array}{c}\vdots\\ \pi_1'\\ \vdots\\ \Gamma\Delta, \exists x\, B(x), B\end{array}}{\Gamma\Delta, \exists x\, B(x)}$$

with $\operatorname{ht}\pi' \leq \operatorname{ht}\pi$. □